

(j) Co-ordination and response	
Students should:	
2.80	understand how organisms are able to respond to changes in their environment
2.81	understand that homeostasis is the maintenance of a constant internal environment, and that body water content and body temperature are both examples of homeostasis
2.82	understand that a co-ordinated response requires a stimulus, a receptor and an effector
<i>Flowering plants</i>	
2.83	understand that plants respond to stimuli
2.84	describe the geotropic and phototropic responses of roots and stems
2.85	understand the role of auxin in the phototropic response of stems

Students should:	
<i>Humans</i>	
2.86	describe how nervous and hormonal communication control responses and understand the differences between the two systems
2.87	understand that the central nervous system consists of the brain and spinal cord and is linked to sense organs by nerves
2.88	understand that stimulation of receptors in the sense organs sends electrical impulses along nerves into and out of the central nervous system, resulting in rapid responses
2.89	understand the role of neurotransmitters at synapses
2.90	describe the structure and functioning of a simple reflex arc illustrated by the withdrawal of a finger from a hot object
2.91	describe the structure and function of the eye as a receptor
2.92	understand the function of the eye in focusing on near and distant objects, and in responding to changes in light intensity
2.93	describe the role of the skin in temperature regulation, with reference to sweating, vasoconstriction and vasodilation
2.94	understand the sources, roles and effects of the following hormones: adrenaline, insulin, testosterone, progesterone and oestrogen
2.95B understand the sources, roles and effects of the following hormones: ADH, FSH and LH	